

SOLVING CONTEXT SHUFFLE WITH OPENSEARCH

MITIGATING PERMUTATION-INDUCED HALLUCINATIONS IN PRODUCTION

Ramya Ravi

 [@ramya-ravi19](#)

 [@ramya_ravi1](#)

 [@ramya_ravi19](#)

David vonThenen

     [@davidvonthenen](#)

INTRODUCTIONS

Ramya Ravi



- **Developer Advocate - AI and Open Source**
- **WiDS Ambassador**
- **AI Books Technical Reviewer for BPB and Apress Publications**

 [@ramya-ravi19](#)
 [@ramya_ravi1](#)
 [@ramya_ravi19](#)

David vonThenen



- **Agentic Solutions**
- **Book In Progress**
- **AI Harness Wizard**
- **NLP Sorcery**
- **AI/ML Security Hacker**
- **Part-time GPU Owner**

     [@davidvonthenen](#)

AGENDA

- **Research Paper: Use Prompt Repetition?**
- **Deconstructing RAG Prompts**
- **Discuss: "Adversarial Prompting"**
- **Prompt Evaluation**
- **Q & A**

RESEARCH PAPER DISCUSSION

PROMPT REPETITION IMPROVES NON-REASONING LLMs

THE RESEARCH PAPER...

Link:
<https://arxiv.org/abs/2512.14982>



 Search [Submit](#) [Donate](#) | [Log in](#)

Computer Science > Machine Learning

[Submitted on 17 Dec 2025]

Prompt Repetition Improves Non-Reasoning LLMs

[Yaniv Leviathan](#), [Matan Kalman](#), [Yossi Matias](#)

When not using reasoning, repeating the input prompt improves performance for popular models (Gemini, GPT, Claude, and Deepseek) without increasing the number of generated tokens or latency.

Subjects: **Machine Learning (cs.LG)**; Artificial Intelligence (cs.AI); Computation and Language (cs.CL)

Cite as: [arXiv:2512.14982](#) [cs.LG]

(or [arXiv:2512.14982v1](#) [cs.LG] for this version)

<https://doi.org/10.48550/arXiv.2512.14982> 

Submission history

From: Matan Kalman [[view email](#)]

[v1] Wed, 17 Dec 2025 00:33:51 UTC (1,403 KB)

Access Paper:

[View PDF](#)

[HTML \(experimental\)](#)

[TeX Source](#)



[view license](#)

Current browse context:
cs.LG

[< prev](#) | [next >](#)

[new](#) | [recent](#) | [2025-12](#)

Change to browse by:

[cs](#)

[cs.AI](#)

[cs.CL](#)

References & Citations

[NASA ADS](#)

[Google Scholar](#)

[Semantic Scholar](#)

THE PROMPT...

MiddleMatch Here the model gets a list of N names or numbers (out of a total possible set of K , i.e., $K < N$ means there will be repeating elements), and is asked to output the name/number located directly between two given ones. We use $N = 40, K = 10$. Example:

Here's a list (potentially with repetitions) of names:

Carlos Davis, Dale Sims, Carlos Davis, Dale Sims, Stephen Cruz, Dale Sims, Finnian Ross, Stephen Cruz, Stephen Cruz, Gregory Collins, Dale Sims, Stephen Cruz, Carlos Davis, Stephen Cruz, Dale Sims, Dale Sims, Stephen Cruz, Stephen Cruz, Leonard Kalman, Bruce Phillips, Raymond Roberts, Dale White, Leonard Kalman, Finnian Ross, James Wright, Finnian Ross, Raymond Roberts, Dale Sims, Dale Sims, Leonard Kalman, Dale Sims, Carlos Davis, Leonard Kalman, Bruce Phillips, Dale Sims, Raymond Roberts, Gregory Collins, Gregory Collins, Dale Sims, Finnian Ross

What is the single name that appears right between Carlos Davis and Bruce Phillips?

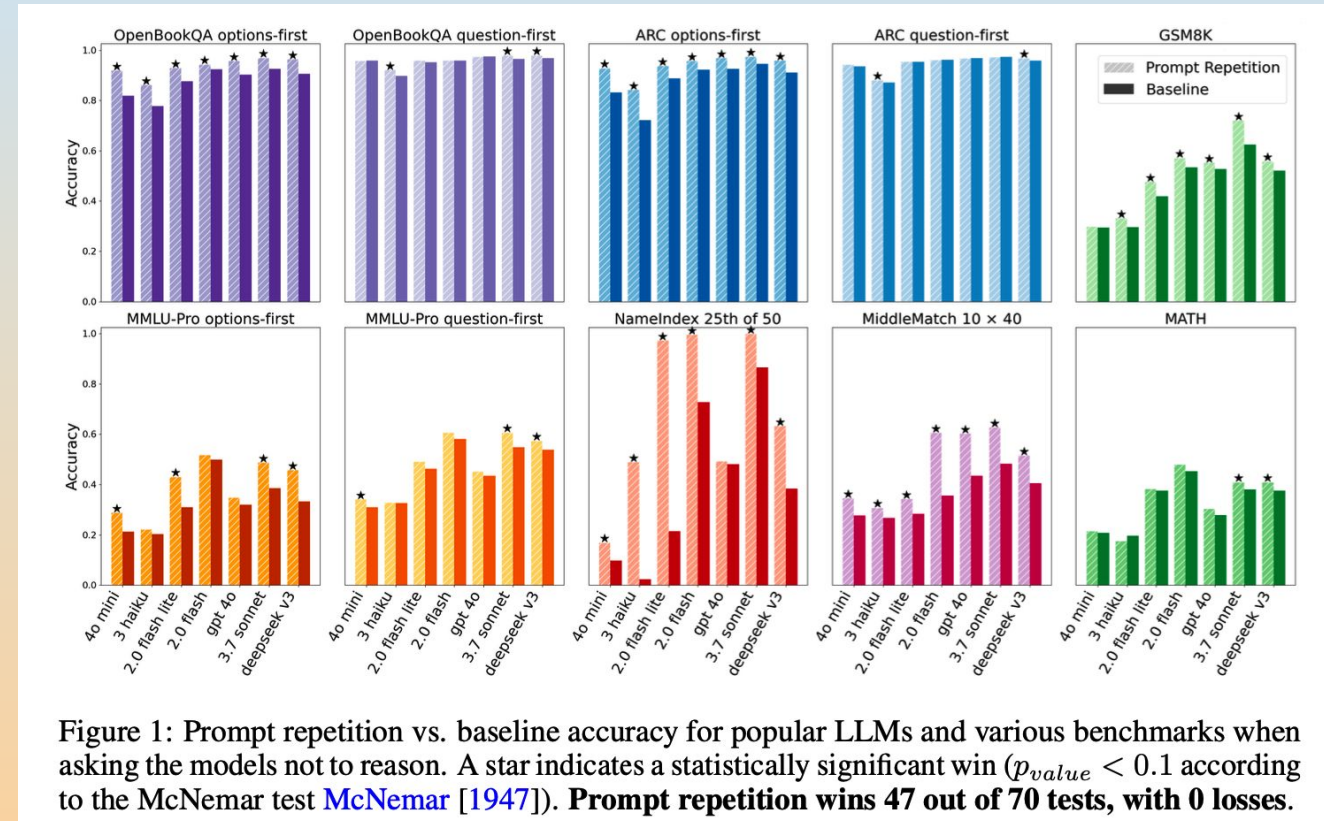
DEMO: PROMPT REPETITION

IN MULTIPLE WAYS...

[HTTPS://YOUTU.BE/zAN2-D5A_dSQ](https://youtu.be/zAN2-D5A_dSQ)

RESULTS

- **Qwen 2.5 7B**
 - Single Prompt: Wrong
 - Duplicate Prompt: Correct
- **Qwen 3.5 9B**
 - Disable Reasoning: Wrong
 - Enable Reasoning: Correct
- **OpenAI**
 - GPT-4o / mini: Not Available 😊
- **Claude**
 - Sonnet 3.7 / Haiku 3: NA 😊



Links:

Qwen 2.5 7B GGUF - <https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-GGUF>

Qwen 2.5 7B MLX (Apple) - <https://huggingface.co/mlx-community/Qwen2.5-7B-Instruct-4bit>

WHY DOES THIS WORK?

- **Language Models (LMs) Are Language Based**

- An (Non-Reasoning) LLM Reads From Left to Right
- Context Follow By Question, What is Relevant? Fail!
 - Prompt Repetition Works, Gives a 2nd Pass
- This is Why Reasoning LLMs Are Unaffected
 - At Cost of (Reasoning) Tokens & Your Wallet!

- **How Does This Affect RAG?**

- Permutation-Induced Hallucinations
- Long Context and Prefill Efficiency

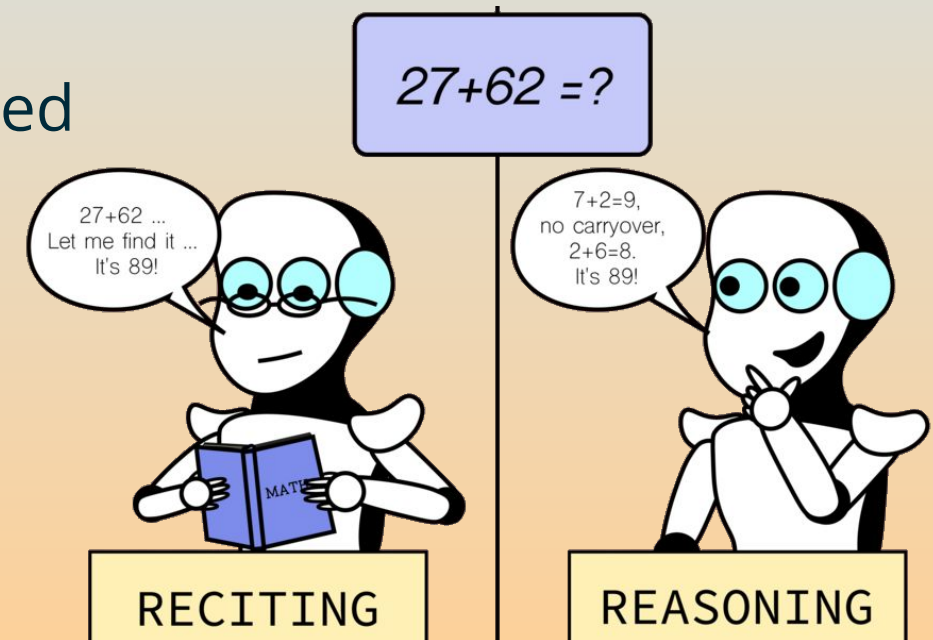


Image Attribution:

Reasoning skills of large language models are often overestimated

<https://news.mit.edu/2024/reasoning-skills-large-language-models-often-overestimated-0711>

RECONSTRUCTING RAG PROMPTS

RESEARCH-BASED PROMPT STRUCTURE FOR RAG

GOOD STRUCTURE VS. BAD STRUCTURE

- **Problem: Cross-Chunk Bleeding**

- Mixing Data without Labels - facts from different sources get blended together
- Fix: Wrap each chunk in a labeled tag
`<chunk id="1" source="Q3 Filing">...</chunk>`
- Now the model knows exactly where each fact came from

- **Problem: Causal Misalignment (Premature Interrogation)**

- Asking the question too early - LLMs read Top-to-Bottom
- Question First = Model starts guessing before it's even seen the facts
- Fix: Always show the Data first, Ask the Question last

DEMO: GOOD STRUCTURE VS. BAD STRUCTURE

Watch the Answer flip from Wrong → Right

[HTTPS://YOUTU.BE/Uq7s09J_SwU](https://youtu.be/Uq7s09J_SwU)

Good Structure Prompt

GOOD STRUCTURE:

- Bounded chunks with explicit id/source delimiters (fixes Cross-Chunk Bleeding)
- Context provided BEFORE the question (fixes Causal Misalignment)

Prompt sent to model:

Retrieved context:

<chunk id="1" source="Apple Inc. 10-Q filing - Q3 FY2025 - Revenue">

Total revenue for the quarter was \$94.9 billion.

</chunk>

<chunk id="2" source="Apple Inc. 10-Q filing - Q4 FY2025 - Revenue">

Total revenue for the quarter was \$102.5 billion.

</chunk>

<chunk id="3" source="Apple Inc. 10-Q filing - Q4 FY2025 - Gross Margin">

Gross margin for the quarter was 47.3 percent.

</chunk>

<chunk id="4" source="Apple Inc. 10-Q filing - Q3 FY2025 - Gross Margin">

Gross margin for the quarter was 46.9 percent.

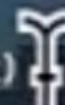
</chunk>

Question: What was Apple's gross margin for the quarter with \$94.9 billion in revenue?

SLM Response (Good Structure):

46.9%

Expected answer: 46.9 percent (the margin attached to the \$94.9B/Q3 chunk)



GOOD STRUCTURE - FROM LLM PROVIDERS

- **OpenAI Cookbook -**

- <https://github.com/openai/openai-cookbook/tree/main/examples>
- Use delimiters to separate data from instructions

- **Anthropic Prompt Engineering Tutorial -**

- <https://github.com/anthropics/prompt-eng-interactive-tutorial>
- Put long context first, Query last
- Use XML tags to structure input - same chunk pattern

"ADVERSARIAL PROMPTS"

**WHAT PROMPT INJECTION TEACHES US ABOUT BAD
STRUCTURE**

WHAT ARE ADVERSARIAL ATTACKS?

- **Intentional Manipulation to Affect LM Outputs**
 - In Security, It's Called Pentesting (Penetration Testing)
 - "How do I Hack This to do XYZ?"
 - With LM, Language is VERY Ambiguous!
- **Basic Examples:**
 - Manipulating Classifiers
 - In Language, [Sentiment Analysis](#)
 - In Visual, [Block Person Identification](#)
- **[WeAreDevelopers Video: Confuse, Obfuscate, Disrupt](#)**
 - Apply Pentesting to Get Results/Output!



EXAMPLE: JSON OR RAW PAYLOAD DUMP

- Remember, LLMs Are **LANGUAGE** Based!

- AI Engineers Get "Lazy" Attach Raw Data
- Using JSON Means Substantial Token Bloat
 - Interfers with Chain-of-Thought

- Research: +5-20% Failure Rates

- How To Fix This? (Platform Specific)

- Heavy Preference for XML
- Alternatives: CSV (not joking), TOON
- Upshot: Use Human-Readable Formats!

License: CC BY-SA 4.0
arXiv:2408.02442v1 [cs.CL] 05 Aug 2024

Let Me Speak Freely? A Study on the Impact of Format Restrictions on Performance of Large Language Models

Zhi Rui Tam
Equal contribution Appier AI Research

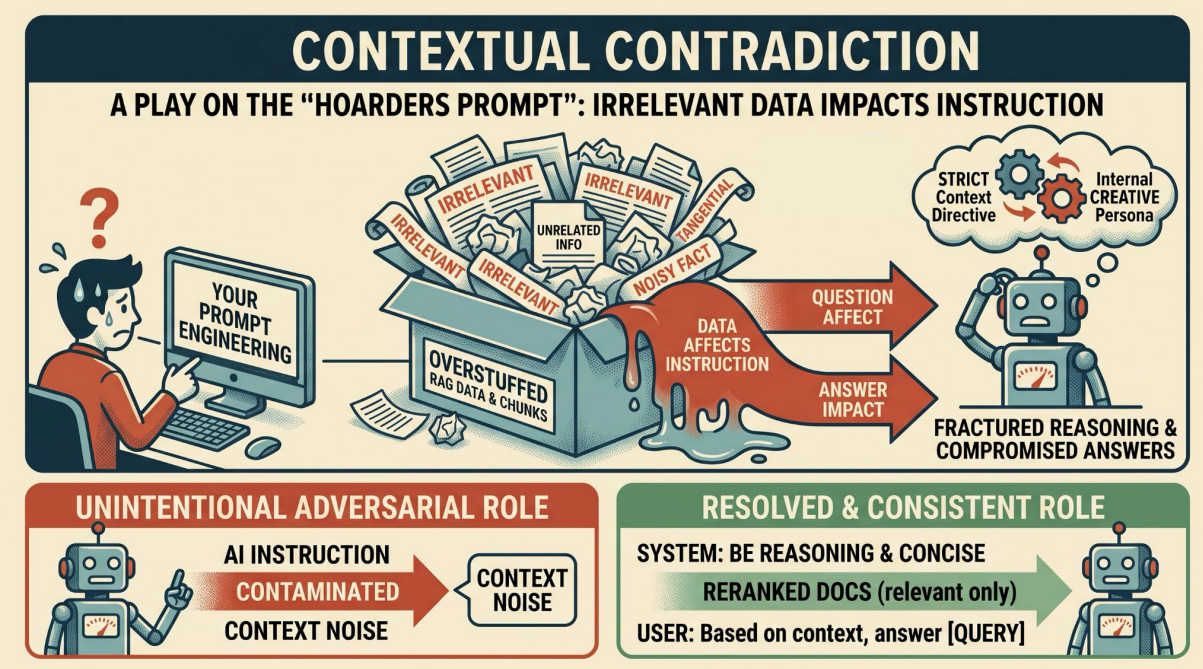
Cheng-Kuang Wu ¹
Appier AI Research

Chieh-Yen Lin
Appier AI Research

Hung-yi Lee
Equal advisorship National Taiwan University

EXAMPLE: CONTEXTUAL CONTRADICTION

- Aka the "Hoarders Prompt"
- How This Happens?
 - Over Retrieving RAG Data
 - Top-K of 10 Might Be Too Much
 - RAG Chunks Are Irrelevant to Question
 - Data Effects Question & Answer
 - Even Irrelevant Data



DEMO: CONTEXTUAL CONTRADICTION

THE BUFFET OF AMBIGUITY

[HTTPS://YOUTU.BE/LCvRBxWiEvI](https://youtu.be/LCvRBxWiEvI)

REVIEW: How We Got Here

- **Poor Data Quality**
 - Hoarders Prompt
 - Contextual Contradiction
- **How to Use Our Results?**
 - Implement ranking drop-off
 - Check for Inconsistencies

Using Adversarial Prompting Challenges Our Prompt Quality

We Want to Avoid Garbage In, Garbage Out



Image Attribution:

GIGO — Garbage in Garbage Out Concept for UX Research

<https://medium.com/nyc-design/gigo-garbage-in-garbage-out-concept-for-ux-research-7e3f50695b82>

PROMPT EVALUATION

How Do You EVALUATE WHAT "GOOD" Is?

Two CHECKS FOR A GOOD ANSWER

- **Faithfulness/Groundedness (LLM-as-Judge)**
 - Ask a second LLM to Grade the first one's answer
 - Question it asks: Does every claim match the source?
 - Industry Terms: RAGAS calls this **Faithfulness**, TruLens calls it **Groundedness**
- **Citation/Attribution Overlap (Heuristic)**
 - What % of the answer's words/numbers actually appear in the source data
 - Catches the #1 hallucination (for example - a number that's nowhere in the context)
 - Industry Terms: Similar to RAGAS Context Precision

We used OpenSearch here. The source data is not hardcoded.

DEMO: PROMPT EVALUATION

Same question, two answers - one good, one bad
Watch both get graded live, side by side

[HTTPS://YOUTU.BE/LQjNjAKUyXA](https://youtu.be/LQjNjAKUyXA)

FOR PRODUCTION : REACH FOR A REAL FRAMEWORK

- **RAGAS - Same faithfulness/precision checks**
- **DeepEval - Built for automated testing in CT pipelines**
- **TruLens - Tracks groundedness across a whole application**

This Demo's checks are a teaching tool!

GALILEO EVALUATION

Not open source, my personal project -
https://github.com/ramyaravi1/health_rag_galileo_eval

TAKEAWAYS

- **Delimiters + Question-last - Zero extra tokens, often the answer flips**
- **Filtering beats hoarding**
- **Evaluate selectively, not on every request**
- **Cost control isn't an afterthought, it's part of the fix**
 - Prompt repetition and reasoning both trade tokens for accuracy
 - Bigger Context is not equal to Better context - More Tokens

SESSION RESOURCES

RESOURCES

SCAN THE QR CODE FOR ALL RESOURCES!!

github.com/instaclustr/code-samples

Announcements:

Launching Instaclustr Labs

github.com/instaclustr.com/labs

Launching Instacademy

<https://www.instaclustr.com/instacademy-courses>



THANK YOU

Ramya Ravi

 [@ramya-ravi19](https://www.linkedin.com/company/ramya-ravi19)
 [@ramya_ravi1](https://github.com/ramya_ravi1)
 [@ramya_ravi19](https://twitter.com/ramya_ravi19)

David vonThenen

     [@davidvonthenen](https://www.linkedin.com/company/davidvonthenen)



Solving Context Shuffle with OpenSearch: Mitigating
Permutation-Induced Hallucinations in Production